Investigating Ad Targeting Differences Between Child and Adult Google Profiles
DRAFT

Muhammad Bilal Ali Aditi Balaji Michael Stoneman

Abstract

As children increasingly access online content, concerns about targeted advertising and exposure to inappropriate content have grown. Although platforms like Google claim to protect minors through compliance with regulations such as COPPA, the effectiveness of these protections remains understudied. This study investigates whether Google's child account protections successfully prevent targeted advertising and adult content from reaching minors. We created three Google profiles—one adult and two child accounts—and compared the advertisements served to each over a two-week period. Our findings indicate that child accounts are served ads that are based on the specific search query, which are contextually appropriate and not based on behavioral targeting. In contrast, the adult account received frequent targeted advertisements based on demographic information. Interestingly, child accounts receive almost no ads when adult terms and age-inappropriate keywords are queried. These results suggest that Google's child account protections are largely effective, though we have identified limitations and areas for future research.

1 Introduction

Online advertising has become increasingly sophisticated, with platforms employing complex behavioral tracking and targeting mechanisms to deliver personalized advertisements. While this personalization may benefit advertisers and adult users, it raises significant concerns when applied to children. The intersection of targeted advertising and children's content consumption poses risks ranging from privacy violations to exposure to age-inappropriate content. Despite regulations such as the Children's Online Privacy Protection Act (COPPA) in the United States which restricts the collection of personal information from children under 13, questions remain about how effectively major platforms implement these protections in practice.

Parents often create supervised child accounts on platforms like Google, trusting that these accounts will shield their children from aggressive ad targeting and inappropriate content. However, previous research has shown that child-directed websites frequently contain trackers and targeted advertisements that may violate privacy regulations. Given the sophistication of modern advertising platforms and the evolving nature of targeting mechanisms, it is crucial to empirically evaluate whether the protections offered by child accounts function as intended.

This study addresses these concerns by conducting a controlled comparison of advertisements shown to adult versus child Google profiles. We investigate two primary research questions:

1. Do child Google profiles receive any targeted ads beyond simple query-based results?
2. How do advertisements differ between adult profiles and child profiles when identical "adult" searches are performed?

To answer these questions, we created three Google profiles: one adult account (A1) and two child accounts (K1, K2). Over a two-week period, we primed these accounts with age-appropriate search behavior to set a baseline, then logged the advertisements served to the adult and child accounts to observe the differences and patterns, if any. Our findings reveal that while the quantity of advertisements served to children is similar to what is shown to adults, Google's child account protections substantially limit demographic and behavior specific targeting towards minors. Additionally, searches for similar adult terms through the A1 and K1 profiles revealed that Google's child account protections largely prevent advertisements from being served when minors search for adult content.

In Section 2, we provide context on online advertising mechanisms, children's privacy regulations, and previous re-

Privacy Policy, Law, and Technology FA25

search on ad targeting. Section 3 describes our experimental design and ethical considerations. We present our results in Section 4 and discuss their implications, limitations, and future directions in Section 5.

2 Background

2.1 Online Ads and Targeting Mechanisms

Online advertising has evolved into an ecosystem that is heavily dependent on user profiling and behavioral targeting. Modern advertising platforms employ both stateful (cookie-based) and stateless (fingerprinting-based) tracking methods to identify and target users across websites [2]. As of 2022, over 90% of websites include at least one tracking script [1]. Platforms use various targeting mechanisms to serve ads to users that they believe will be receptive, including those based on demographics, interests, behaviors, locations, and contexts. These targeting categories can be combined when placing an advertisement, allowing advertisers to reach highly specific audiences.

Google, for instance, provides five broad categories of targeting that advertisers can utilize on YouTube to reach the audience they desire—demographic groups (based on age, gender, parental status, household income); detailed demographics (based on broad shared audience traits such as being a college student, a homeowner or a new parent); interests (where the advertiser can further filter users based on "affinity segments", "custom affinity segments", "life events", "in-market segments", and "custom segments"); data segments (based on past interactions the user has had with the advertiser's YouTube channel), and customer match (a mechanism by which the advertiser can re-target their existing customers). Along with this, Google also offers specific targeting methods—placement-based, topic-based, keyword-based, and device-based. These methods, combined with audience targeting filters, are to help advertisers reach the right audiences [6].

2.2 Children's Exposure to Online Advertising

The intersection of targeted advertising and children's content consumption poses significant concerns. Multiple jurisdictions have established regulations to protect children from inappropriate advertising and invasive targeting. In the United States, the Children's Online Privacy Protection Rule (COPPA) restricts the collection of personal information from children under the age of 13, absent verifiable parental consent. Cookies and fingerprints fall under this restriction, limiting profile-based advertising while still permitting contextual advertising [4].

Ad platforms like Google claim to protect children by not allowing advertisers to target users under the age of 18 through their advertising interface. However, research has shown that other targeting features can be exploited to research children, such as placement-based targeting on videos made for children [4].

2.3 Prior Research

Previous studies have highlighted the pervasive presence of trackers and targeted advertisements on the internet. Moti et al. conducted such a study of child directed websites. They did so by first creating a multilingual classifier to identify such sites from over two million pages in the Common Crawl dataset. Their analysis of 2,000 child-directed websites reveals that around 90% embed one or more trackers and approximately 27% contain targeted advertisements, which should require verifiable parental consent [5]. Khan et al. examine advertisement exposure and content in child-oriented videos, comparing labeled videos across different countries. They find that children's ad exposure depends heavily on regional context and that weak enforcement of Youtube's child protection policies allows inappropriate ads to appear [3].

Medjkoune, Goga and Senechal (2023) had recorded that Google provided seven broad categories of targeting for advertisements on YouTube: location-based, profile-based, interest-based, keyword-based, topic-based, placement-based, and re-targeting [4]. Today, Google's platform offers advertisers with more options to target their desired audience on a more granular level [6]. The evolution of the options advertisers have has made research in this domain increasingly important, as it highlights how personalization and targeting mechanisms are becoming more sophisticated, and potentially more invasive.

3 Methodology

This study explores whether ad platforms like Google's allow online advertisers to (1) target children, and (2) show age-inappropriate or adult ads to children. To do this, we created three profiles with Google: one for an adult (A1) and two for children aged 11-12 (K1, K2). Each profile had its own account, and was tied to a separate Chrome profile, so that Google treated them as separate users. The adult profile represented normal adult browsing behavior, while the two child profiles were used to mimic children's online activity.

One of the child accounts, K1, was used to try to "bait" Google into serving inappropriate ads to a minor. It did so by searching for adult content about marijuana, cigarettes, and vaping. The other child account, K2, was used as a control, to see if any inappropriate ads were shown to children at all, without any search queries prompting such ad serving. We decided on this specific age range owing to the fact that most regulations in the US (and elsewhere in the world) clearly classify such users as minors.

Before setting up the accounts, we deleted all existing Chrome data, including history, cookies, and bookmarks. We then created new Google accounts for each profile and signed

into Chrome. We kept Google's default privacy and ad settings so that ad personalization worked normally. This helped us see how Google's system automatically targeted ads for each type of user.

We spent the first two weeks "priming" each user's profile by searching for generic "age-appropriate" content. The adult searched for normal content, with some searches being for mature topics, and others being generic. The children profiles (K1 and K2) simulated Google and YouTube search activity and YouTube video viewing activities of 11 and 12-year-olds.

Our original plan to allow Google to create a profile for each of these users which we would monitor via the security and privacy dashboard associated with each Google account was limited by the fact that Google did not allow for such metrics to be tracked for the child accounts. Thus, we used insights from A1's dashboard of what it perceived the user's interests to be, to further fine-tune our search queries for A1. Once the baseline behavior of these profiles were established, we moved into the "experimentation" phase.

In this phase of our project, we recorded the advertisements shown to each profile and analyzed them. We meticulously recorded the following data for the ads that were served: URL/Search Query, Ad Platform, Relevance, Destination URL, Advertisement Type, Location/Placement, Product/Service Advertised, Advertiser Name, Ad Content Description, "For Kids" Video (yes/no), Stated Reason For Targeting, and Hypothesized Targeting Factor. A1 and K2 logged the details for 85 such advertisements, and K1 logged the details for 70 such advertisements.

We then carried out overlapping search queries across the various profiles, to compare and contrast how Google and YouTube treat these different profiles. The K1 and K2 accounts had an overlap of 15 queries—these were searches for terms typically associated with children, such as "Mr beast", "Nintendo switch 2 new games" and "Minecraft building ideas",and were split across YouTube and Google platforms.

The A1 and K1 accounts had an overlap of 15 queries—these were searches for mature terms, such as "Weed gummies near me", "How to roll a joint", and "Is vaping bad for health", and were split across YouTube and Google platforms. If an advertisement was shown at all, we logged it, and if there was no advertisement served (as observed with the K1 account), we noted that down as well as a part of the logs.

Thus, in total, we had 100 logs per profile, and these were analyzed in order to:

1. Compare the ads shown to A1 and K1 for the mature terms and identify the protective measures in place.
2. Compare the ads shown to K1 and K2, and analyze whether the ads shown were targeted or merely based on the search query.
3. Check if Google showed any inappropriate ads to children, even if the child did not specifically look for adult content (K2).

Throughout this experiment, we hoped to confirm whether advertisers/advertising platforms correctly separated adult and child audiences online, and identify whether children were being shown ads unsuitable for them (based on their age).

3.1 Ethics

This study involved analyzing the types of advertisements shown, based on the specific user profile. These user profiles did not use any information about actual people, and any similarity between the names/ages of these profiles and actual people, if any, was purely coincidental. We only visited sites (via the advertisements) that were public and open to everyone, and did not look for anything that would be construed as criminal under Pennsylvania's laws or Federal law.

3.1.1 Stakeholders

Stakeholders that may be affected by this research, and the effects upon them, are as follows:

- **Recipients of online advertisements:** Following the publication of this research, the recipients of targeted advertisements (children and parents) may receive different types of advertisements than they had previously. We expect any change will be for the better, i.e., children will receive fewer inappropriate advertisements than they did before.
- **Ad-tech platforms and advertisers:** The providers of targeted advertisements (advertising companies) may need to change the techniques through which they identify minors and select what ads to serve them.
- **Research Team:** We, the research team, are unlikely to be significantly impacted.

3.1.2 Impacts

We did not use any real person's details, or real person's behavior to carry out our experiment. We tried to simulate how children may naturally come across inappropriate content, without any individual's personal information being used.

We followed Google's rules and process when it came to setting up child accounts, and as required, set it up through a verified parental account.

3.1.3 Harms

We did not identify any harms to come out of this experiment for children or their parents. Advertisers and advertising platforms might have to take a deeper look at their compliance with children's laws, such as COPPA, based on some of the interesting results that were generated via the experiment.

3.1.4 Mitigations

If we find that advertisers/advertising platforms are violating children's privacy laws such as COPPA, we will inform the advertising platform (here, Google) first, and allow them time to take corrective action. If we do not hear from them, we will look at other legal mechanisms to set right this violation, such as reporting our findings to the FTC.

3.1.5 Decision

Our motivation behind this experiment was to analyze the advertisements and content that children are exposed to, as children are a vulnerable group. We decided to have a child profile (K1) that specifically looked for adult content because it mimics something that might possibly occur in reality

4 Results

Our analysis reveals differences in handling of advertisements around mature topics between adult and child Google profiles. These differences were consistent across the data collection period and were particularly pronounced when profiles searched for adult-oriented content.

4.1 Advertisement Frequency and Targeting

Adult Profile (A1): The adult profile received advertisements on approximately half of its adult-oriented searches. These ads fell into two distinct categories based on the search content. For searches unrelated to cigarettes and vaping, advertisements were contextually relevant to the query—for example, a search for "engagement rings near me" triggered ads directly related to jewelry and engagement rings from local retailers. However, for searches related to cigarettes or vaping, no product-related ads appeared. Instead, A1 received advertisements that appeared to be targeted based on demographic information, time of day, or location, such as ads for internet service providers or insurance products.

Google's Privacy & Personalization dashboard for A1 revealed a detailed demographic profile. The platform had inferred that the user was single, high-income, possessed a high school diploma, worked for a very large employer, and was not a parent. This demographic profiling enabled the sophisticated targeting observed in A1's ad experience.

Child Profile K1 (Mixed Browsing): K1, which searched for both child-appropriate content and adult topics (specifically smoking, vaping, and e-cigarettes), showed a markedly different pattern. Across 15 adult search terms, K1 received only a single advertisement, which was unrelated to the search query and appeared to be contextually based on child-appropriate content rather than the adult search term itself.

For non-adult search terms, K1's advertising pattern closely resembled that of K2, receiving occasional contextually appropriate advertisements. Notably, some search results for adult terms appeared visually similar to advertisements but were not marked as sponsored content or ads. Additionally, certain YouTube videos that appeared in Google Search results for adult queries were inaccessible through the K1 profile, indicating that Google Search and YouTube operate with some degree of independence in their content filtering systems.

Child Profile K2 (Control): K2, which searched exclusively for child-appropriate content, did not receive targeted advertisements at any point during the study. However, K2 did see advertisements that appeared designed for children, such as toy-related ads and kid-friendly product promotions. These ads were contextually relevant to the content being viewed rather than based on behavioral profiling.

Additionally, while searching for child-related content on Google, in addition to child-related ads, K2 also saw advertisements, and Call to Actions (CTAs), that can be considered a bit more adult-oriented, such as insurance. /

The following are some of such pages advertised to K2:

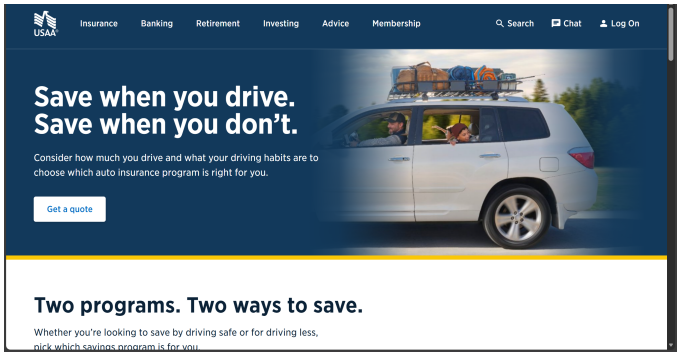


Figure 1: Advertisement 1 For K2

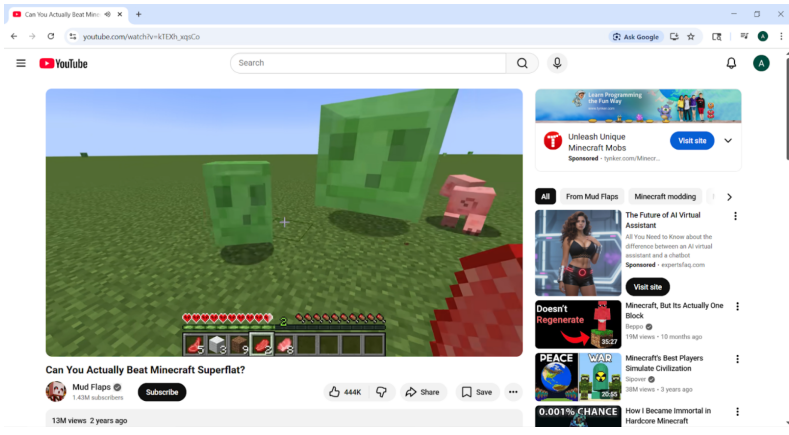


Figure 2: CTA 1 For K2

4.2 Google's Profiling and Account Protections

Examination of Google's Privacy & Personalization dashboards revealed a critical difference in how the platform treats child versus adult accounts. While A1's dashboard contained detailed demographic inferences and interest categories, both K1 and K2's dashboards remained blank throughout the study period. This aligns with COPPA's restrictions on collecting personal information from children under 13 without verifiable parental consent.

Our investigation also uncovered several technical mechanisms Google employs to protect child accounts:

- Google automatically appends a special key-value pair to searches conducted through child profiles. This parameter does not appear in searches from adult profiles, suggesting it serves as an internal flag for content filtering and ad serving decisions.
- Chrome profiles associated with child accounts have incognito mode disabled, preventing children from browsing without parental oversight.
- Google sends automated emails to parents or guardians 30 days before a child turns 13, reminding them that the child will soon be able to make independent choices regarding their Google account settings.

4.3 Search Result Accessibility

An interesting finding emerged regarding content accessibility across platforms. Several YouTube videos returned as Google Search results for the K1 profile were blocked on the YouTube level and unable to be played due to the account's age. An identical search on YouTube did not provide links to said videos, which indicates that while Google Search and YouTube share some safety infrastructure, they also maintain independent filtering systems that may not always be perfectly synchronized.

5 Discussion

Our findings suggest that Google's child account protections are largely effective in preventing both targeted advertising and exposure to inappropriate ad content for users under 13. These results were somewhat surprising, as we had anticipated that the protections might not be as robust as Google's stated policies suggest. The near-complete absence of advertisements during adult searches on the K1 profile, combined with the blank demographic profiles for both child accounts, indicates meaningful implementation of COPPA compliance measures.

5.1 Implications for Stakeholders

For Parents: Our research provides evidence that parents can be moderately more confident in allowing their children to use supervised Google accounts than they might have been without this information. The platform does appear to implement meaningful safeguards against targeted advertising and adult content exposure. However, this should not eliminate the need for parental supervision entirely. Children still see advertisements on these platforms—they are simply contextual rather than targeted—and content filtering, while effective, is not perfect as evidenced by the YouTube videos that appeared in search results but were blocked upon access.

For Google and COPPA Compliance: Based on our observations, Google appears to be in compliance with COPPA's requirements. The platform does not build behavioral profiles of child users, does not serve targeted advertisements to them, and implements technical measures to flag and handle child accounts differently from adult accounts. The automatic addition of special parameters to child account searches demonstrates a systematic approach to differentiation rather than ad-hoc filtering.

For Future Research: Our study opens several avenues for continued investigation. The distinction between Google's stated advertising policies (which prohibit ads for products designed to simulate tobacco smoking and impose harsh restrictions on THC-related advertising) and child account protections remains unclear [7]. It is difficult to determine from our results alone whether K1's lack of vaping and marijuana-related ads stemmed from these general policy restrictions or from child-specific protections. Future research should investigate adult content categories that do not fall under Google's restricted advertising policies to better isolate the effect of child account protections specifically.

5.2 Limitations

Several limitations constrain the generalizability and completeness of our findings. First, we were limited in our ability to collect large amounts of data by the fact that we manually logged every advertisement rather than employing automated data collection tools. This manual approach, while ensuring accuracy, restricted the scale of our study and may have introduced recording inconsistencies.

Second, as noted above, the overlap between Google's general advertising restrictions on tobacco-simulating products and THC-related content, and the child-specific protections makes it difficult to determine the precise source of the advertising patterns we observed. A more comprehensive study using adult content categories without such advertising restrictions would help clarify this issue.

Third, our study examined only Google's ecosystem. Other platforms may implement child protections differently, and our findings should not be generalized beyond Google without

additional research.

Finally, our study period was limited to two weeks of priming and one week of data collection. Longer-term observation might reveal different patterns or edge cases not captured in our timeframe.

5.3 Future Directions

If we were to continue this research, several methodological improvements would strengthen our findings. First, we would develop or employ a Chrome extension to automatically collect and categorize advertisements, enabling larger-scale data collection and reducing manual recording errors. Second, we would expand the range of adult topics searched beyond tobacco and marijuana to include content that does not fall under Google's restricted advertising categories, such as alcohol, gambling, or mature entertainment. This would better isolate the effects of child account protections from general platform policies.

Most significantly, we would introduce a fourth profile: an adult account (A2) that follows the same search pattern as K1—primed with child-appropriate searches and then shifting to adult content. This control would allow us to better determine what differences are caused explicitly by the account's designated age versus the search history and behavioral patterns. For instance, such a profile would help us understand whether an adult account exhibiting child-like behavior would receive fewer ads, or whether the child account designation is the primary protective mechanism.

5.4 Conclusion

Despite growing concerns about children's exposure to targeted advertising online, our study provides evidence that major platforms can implement effective protections when properly motivated by regulation and policy. Google's child account system appears to meaningfully restrict advertisers from serving inappropriate advertisements to minors, even when those minors actively search for adult content. While these protections are not perfect and should not replace parental supervision, they represent a significant improvement over unfiltered exposure to the modern advertising ecosystem. As targeting mechanisms continue to evolve in sophistication, continued research and vigilance will be necessary to ensure that child protections keep pace with industry developments.

References

[1] Savino Dambra, Iskander Sanchez-Rola, Leyla Bilge, and Davide Balzarotti. When sally met trackers: Web tracking from the users' perspective. In *31st USENIX Security Symposium (USENIX Security 22)*, pages 2189–2206, Boston, MA, August 2022. USENIX Association. URL: https://www.usenix.org/conference/usenixsecurity22/presentation/dambra.

[2] Steven Englehardt and Arvind Narayanan. Online tracking: A 1-million-site measurement and analysis. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, page 1388–1401, Vienna Austria, October 2016. ACM. URL: https://dl.acm.org/doi/10.1145/2976749.2978313, doi:10.1145/2976749.2978313.

[3] Emaan Bilal Khan, Nida Tanveer, Aima Shahid, Mohammad Jaffer Iqbal, Haashim Ali Mirza, Armish Javed, Ihsan Ayyub Qazi, and Zafar Ayyub Qazi. Analyzing ad exposure and content in child-oriented videos on youtube. In *Proceedings of the ACM Web Conference 2024*, WWW '24, page 1215–1226, New York, NY, USA, 2024. Association for Computing Machinery. doi:10.1145/3589334.3645585.

[4] Tinhinane Medjkoune, Oana Goga, and Juliette Senechal. Marketing to children through online targeted advertising: Targeting mechanisms and legal aspects. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, page 180–194, Copenhagen Denmark, November 2023. ACM. URL: https://dl.acm.org/doi/10.1145/3576915.3623172, doi:10.1145/3576915.3623172.

[5] Zahra Moti, Asuman Senol, Hamid Bostani, Frederik Zuiderveen Borgesius, Veelasha Moonsamy, Arunesh Mathur, and Gunes Acar. Targeted and troublesome: Tracking and advertising on children's websites. In *2024 IEEE Symposium on Security and Privacy (SP)*, pages 1517–1535, 2024. doi:10.1109/SP54263.2024.00118.

[6] Google Support. About targeting for video campaigns. URL: https://support.google.com/youtube/answer/2454017.

[7] Google Support. Dangerous products or services. URL: https://support.google.com/adspolicy/answer/6014299.